

VISHNU SHARMA

AI Solutions Architect | Founder, Prompt Pulse | Full-Stack Developer

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PROFESSIONAL PROFILE

AI Solutions Architect and Founder of Prompt Pulse specializing in autonomous, multi-agent systems and full-stack web architectures. Designer and sole engineer of TRON-X, a self-hosted personal AI platform spanning 30,000+ lines of Python, 240+ REST endpoints, and a 106-model intelligent router. Experienced across LLM orchestration, RAG, and hybrid-cloud deployment, with elite competitive-programming rankings on AtCoder and CodeChef.

WORK EXPERIENCE

Founder & Chief Architect — Prompt Pulse (AI Solutions Startup)

Sept 2025 – Present

- Built autonomous agentic systems focused on prompt-to-code generation and secure enterprise reasoning workflows.
- Architected scalable, model-agnostic AI pipelines that automate technical decision-making and task execution.

Web Development Intern — K.S.R.M. College of Engineering

Mar 2025 – Present

- Engineered the official CSE Department Portal with dynamic content management and modular navigation.
- Built and optimized a hybrid Alumni Management System for community engagement; gathered requirements directly from stakeholders.

Backend Development Intern — SKRIBE

May 2025 – Aug 2025

- Developed REST APIs and backend architecture for high-performance document-processing features.
- Implemented secure backend logic and optimized data handling, with comprehensive debugging across the stack.

KEY TECHNICAL PROJECTS

TRON-X — Self-Hosted Autonomous AI Assistant

Sept 2025 – Present

Personal AI Infrastructure — sole architect & engineer

- Architected a self-hosted, multi-agent AI platform in FastAPI spanning 30,000+ lines of Python across 128 modules and 240+ REST endpoints, exposing chat, voice, memory, system control, IoT, calendar, email, and analytics domains.
- Built a SmartRouter over LiteLLM that intelligently routes natural-language requests across 106 models from 14 providers, with circuit-breaking, automatic fallbacks, A/B testing, and per-model latency tracking for fault tolerance.
- Implemented a full RAG pipeline using ChromaDB and SentenceTransformers embeddings with MMR reranking and episodic-memory extraction, enabling context-aware, persistent interactions across sessions.
- Engineered a fleet of autonomous sub-agents (research, browser, code, vision, calendar, email, scheduler, WhatsApp) coordinated by a task decomposer for parallel and sequential multi-step execution; added a sandboxed Python/Bash executor and whitelisted PowerShell automation.
- Integrated Home Assistant + MQTT IoT control and cascading voice I/O (Groq Whisper STT, local Kokoro-ONNX TTS) achieving 1–2 s time-to-first-audio.

Tech Stack: FastAPI, LiteLLM, ChromaDB, SentenceTransformers, Playwright, APScheduler, Docker, systemd, Python, Bash

Academic E-Commerce Engine

Mar 2025 – Apr 2025

Design Thinking & Innovation — UI/UX commerce prototype

- Built a responsive e-commerce prototype in HTML5, CSS3, and vanilla JavaScript with product discovery, cart, checkout, filtering, search autocomplete, wish-list, and a recommendations module; testing showed a 4.8/5 satisfaction score and a 28% drop in cart abandonment.

Tech Stack: HTML5, CSS3, JavaScript (ES6+), Responsive Design, UX/UI

TECHNICAL EXPERTISE

Languages & Core: Python, JavaScript (ES6+), SQL, HTML5, CSS3, REST API design, FastAPI

AI & ML: Multi-Agent Systems, RAG, Prompt Engineering, Generative AI, NLP, LLM Orchestration, ReAct

Tools & Infra: Git/GitHub, Docker, systemd, Linux, Microsoft Azure, ChromaDB, LiteLLM, Playwright, Postman

HONORS & LEADERSHIP

- Global Rank 1**, Polaris AI Programming Contest 2026 (AtCoder ABC 457); **Rank 2**, ABC 456; **Rank 3**, ARC 218; **Rank 10 & 76**, CodeChef Starters — top 0.5% worldwide.
- Rank 107**, LeetCode Biweekly 182; Perfect Score (500/500), UOI 2026 Day 2. **President**, CSI — KSRMCE Chapter; **Student President**, KSRMCE Alumni Association.

EDUCATION

B.Tech, Computer Science & Engineering — K.S.R.M. College of Engineering (Autonomous), Kadapa

2023 – 2027

Intermediate (TSBIE) — Deeksha Junior College, Hyderabad

2020 – 2022

SSC, 10th Class — Bhashyam High School, Hyderabad (CGPA 9.3)

2020